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Evaluation Model for a Port Hinterland Intermodal Freight Network Considering Environmental Impacts and Capacity Constraints

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Abstract

The huge demand for containerized cargo and the low market share of rail and inland waterway transport in the hinterland freight system in China are causing severe traffic congestion and pollution. Therefore, this paper focuses on innovative modeling to evaluate the effects of transport policies on the reduction of carbon emissions and the mode shift from road to low-carbon modes in a three-mode port hinterland freight network. The proposed model can capture the main characteristics of the freight system such as mode transfer at inland terminals, transport capacity and processing capacity constraints, flow-dependent link performance functions, and shippers' perceptual error of generalized transport costs. The model assumes that shippers choose transport route, mode, and inland terminal at the same time in a user equilibrium manner, and that bundles of container flow passing competing ports follow the logit formulation. Computational results based on the freight network in the Yangtze River Economic Belt in China indicate that carbon tax, intermodal transport subsidy, and capacity expansion policies can reduce total carbon emissions and promote mode shift from road transport to rail and inland waterway transport, and policy packages show better network performance compared with a single policy type. It is noted that transport policies sometimes lead to the paradoxical phenomenon. Finally, sensitivity analyses are carried out on parameters of cost, time, carbon emissions, and error to test the robustness of the model.

Keyword

port hinterland intermodal freight network, transport policies, user equilibrium, capacity constraint, mode shift

Over the decades, interregional economic cooperation and competition have led to a significant growth in international trade (*1*). Maritime transport plays a vital role in global freight transport, with about 86% of the world's merchandise trade (reaching 10.7 billion tons in 2020) being transported via shipping. Seaports typically serve as gateways for both import and export cargo, and the hinterland freight system, which connects seaports to customers' locations, accounts for about 80% of the total transport costs (*2*). However, the growth in the number of containers in most countries has not matched infrastructure construction and transport policy formulation for the hinterland freight system (*3*). Consequently, the volume of road freight has increased significantly, and excessive use of the road network produces serious externalities, including traffic congestion and air pollution. In this context, intermodal transport emerges and acts as an alternative to road transport (*4*). Intermodal transport refers to sequential transport through at least two modes of transport, with mode transfer being performed at an inland terminal. Compared with single-mode transport, the flexibility provided by mode shift in intermodal transport offers a way of reducing environmental impacts.

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Since the economic reform and opening up, China's freight industry has experienced tremendous growth, with freight volume rising from 3.19 billion tons in 1978 to 52.1 billion tons in 2021. Seaport development has been particularly remarkable, with about half of the world's top 20 container seaports now located in China. However, in the port hinterland transport system, road transport has a high market share, with the average freight distance increasing from 70 km to 180 km, whereas the development of intermodal transport such as rail and inland waterway is lagging behind (*5*). Specifically, as the largest seaport in China with regard to rail traffic, the Port of Ningbo-Zhoushan transported 1.2 million TEUs (20-ft equivalent unit) in 2021, which accounted for only 3.86% of the port's container throughput, whereas road transport volume surpasses 70%. The Port of Shanghai's road transport share is about 50%, and the port has had the largest annual container throughput in the world for 12 consecutive years. To facilitate mode shift from road to intermodal transport, the Chinese government has signed numerous policy documents and put forward various development plans, for example, the three-year action plan to promote adjustments to the transport structure (2018–2020) and the work plan to promote the development of intermodal transport to optimize the transport structure (2021–2025). In particular, with the Chinese government's goal of "reaching the carbon peak by 2030 and carbon neutrality by 2060," intermodal transport has entered a new phase and will play an important role in restructuring and optimizing the transport structure. Policy measures taken so far are infrastructure construction and freight subsidies; internalization of the external costs of transport through imposing a carbon tax is another policy still under discussion by the government and the public.

This paper aims to evaluate the effects of different policies on reducing carbon emissions and promoting mode shift through quantitative modeling of port hinterland intermodal freight networks, to provide decision-making support for the government and assist in formulating transport policies. Specifically, the network consists of a road network, a rail network, and a inland waterway network, and they are connected at inland terminals. Shippers select seaports based on generalized transport costs, including link transport costs, time costs, carbon emissions costs, and node operational costs. Because transport time and carbon emissions are related to vehicle type, driving speed, traffic conditions, node operational equipment, and other factors, there will be some deviation between the shipper's estimated time and carbon emissions, and the actual values. Therefore, generalized transport costs perceived by the shipper are the sum of actual costs and perceptual error, and the model recognizes the generalized costs perceptual error by applying the logit model to determine the bundles of container flows passing competing seaports. Furthermore, the model explicitly considers model transport capacity and inland terminal processing capacity constraints, and incorporates traffic congestion on the road, rail, and waterway links using a user equilibrium traffic assignment technique (*6*, *7*). As such, our approach allows for both a complete investigation of the relationship between infrastructure capacity supply and demand and a clearer understanding of the interaction between link performance and link flows. Moreover, the impacts of transport policies, including mode and inland terminal capacity expansion, a carbon tax, and subsidies for rail and inland waterway transport can be tested by this model.

The remainder of the paper is organized as follows. First, the port hinterland freight network, mathematical model, and algorithm design are formally described. Then, a review of the relevant literature is provided. Next, the model is applied to a case study in China and the effects of transport policies are explored. Finally, conclusions and suggestions for further research are presented.

Literature Review

Intermodal freight network optimization has been investigated extensively (*8–12*), and the objective is to determine transport routes and container flow distribution to minimize transport costs or time. In the last few years, the literature on port hinterland network optimization has continued to increase. Iannone (*13*) built a mathematical model for the strategic planning of port hinterland freight transport, mainly to measure the advantages of employing intermodal solutions. Cho et al. (*14*) provided a weighted constrained shortest path model and used a dynamic programming method to find Pareto optimal solutions to minimize total costs and total time. Li et al. (*15*) proposed a receding horizon container flow assignment approach to address the intermodal freight-planning problem, considering dynamic transport demands. It should be noted that multiple modes of transport are available, and the inland terminal connects the road network with the rail/waterway network. Therefore, some papers consider the construction of inland terminals when resolving network optimization problems. Halim et al. (*1*) utilized a multiobjective optimization model to select the locations of distribution centers that could recognize company service level preference. Teye et al. (*3*) introduced an entropy maximizing model to find the optimal terminal location, and observed and unobserved factors were allowed to affect the shipper's choice behavior. The above models expressly discuss the topological structure of transport networks. However, these studies neglect the influence of

traffic congestion on path choice. The network equilibrium assignment model can remove this restriction by adopting the flow-dependent link performance function, which represents the actual travel time on a link as a function of link flow. For example, Yamada et al. (*16*) built a bi-level programming model to design freight networks in which a multiclass user equilibrium traffic assignment technique was incorporated within the lower-level problem. Uddin and Huynh (*17*) developed a model for equilibrium traffic assignment in large-scale road–rail intermodal networks. In practice, transport capacity in the network is limited, and flow assignment is subject to link and node capacity constraints. However, there has been little effort to study the problem of combining equilibrium flow assignment and capacity constraints for port hinterland freight networks (*18*, *19*). Moreover, conventional solutions to port hinterland freight network problems mainly assign flows to routes based on generalized costs and do not recognize variable perception of costs, which indicates there are also some perceptual errors in relation to transport costs that affect the assignment of freight flows.

Extensive research has been conducted on developing reasonable methods for assessing the impact of policy measures on the port hinterland freight system. Macharis et al. (*20*) developed a LAMBIT model for verifying the influence of different fuel prices on the road transport market share. Santos et al. (*21*) delimited potential hinterlands of terminals according to the generalized costs and suggested that container terminal relocation had an impact on hinterland geography. These models mainly concentrate on pure cost minimization, and do not involve environmental costs. Environmental protection is now a hot topic worldwide, and the reduction of emissions in the transport industry has attracted widespread attention from government and academia (*22*). The conflict between economic goals and ecological goals is an important feature in freight-planning activities. Zhang et al. (*23*) built a mathematical programming model that merged intermodal infrastructure and service network in which a carbon tax and terminal construction scheme were chosen as measures. Dai et al. (*24*) exploited an integrated decision framework to compare and analyze emissions limitations, a carbon tax, and carbon trading policies. Carboni and Orsini (*25*) presented a traffic assignment model to calculate the total reduction in carbon emissions when adding a dry port to the freight network. A review of port hinterland transport and logistics has been compiled by Behdani et al. (*26*), who considered sustainable development would be the focus of future research. However, these models mainly utilize market-based tools to internalize carbon emission costs, including a carbon tax or carbon trading, and do not discuss the effects of subsidies on mode shift and carbon reduction, given that they are usually employed to cut down traffic congestion or environmental pollution in the transport sector (*27*). In point of fact, none of these models can evaluate infrastructure capacity, transport planning, and multiple green regulatory policies for the port hinterland intermodal network simultaneously.

The relevant literature is summarized in Table 1. In conclusion, this paper intends to fill the gaps identified by building a nonlinear integer programming model in a three-mode port hinterland intermodal freight network, considering capacity constraints, perceptual error in relation to cost, and flow-dependent link performance function, to analyze the effects of transport capacity expansion, a carbon tax, and subsidy policies on mode shift, carbon emissions, and network costs.

Table 1. Summary of Port-Hinterland Relevant Papers in the Literature

Reference number	Traffic congestion	Capacity constraint	Cost perception error	Environmental costs	Type of Mode
1	na	√	na	na	One
3	na	√	na	na	Two
13	na	√	na	na	Two
14	na	na	na	na	Three
15	na	√	na	na	Three
16	√	na	na	na	Two
17	√	na	na	na	Two
18	√	√	na	na	Three
19	√	√	na	na	Three
20	na	na	na	na	Three
21	na	na	na	na	Two
23	na	na	na	√	Three
24	na	√	na	√	Three
25	na	na	na	√	Two
Current research	√	√	√	√	Three

Note: na= not applicable.

Table 2. The Sets, Constants, and Decision Variables in the Model

Sets	
I	Set of inland cities, $i \in I$
J	Set of seaports, $j \in J$
S	Set of inland terminals, $s \in S$
D	Set of inland terminal virtual nodes, $d \in D$
K_{ij}	Set of paths connecting every pair of cities and seaports
G	Set of road links, $g \in G$
R	Set of rail links, $r \in R$
W	Set of waterway links, $w \in W$
Y	Set of processing links for inland terminals, $y \in Y$
$\overline{A}$	Set of links with capacity constraints, $\overline{A} \subseteq A$
Constants	
θ	The deviation parameter when the shipper chooses the seaport; this is related to the distribution of the shipper's perceptual error of generalized transport costs
α_{ka}^{ij}	Binary variable for the relationship between path and link; if this is 1, the link a is on the path k connecting the city i and the seaport j
P_i	Container freight demand for the inland city i (TEU)
C_a	Transport capacity for the link a (TEU), $a \in G \cup R \cup W$
C_y	Processing capacity for the inland terminal y (TEU)
b_y	Handling cost for the inland terminal s ($/TEU)
b_a	Transport costs for the link a ($/TEU), $a \in G \cup R \cup W$
b_j	Operational costs for the seaport j ($/TEU)
e_y	Carbon emissions rate of the inland terminal y (kg/TEU)
e_a	Carbon emissions rate of the link a (kg/TEU), $a \in G \cup R \cup W$
e_j	Carbon emissions rate of the seaport j (kg/TEU)
t_a^0	Free-flow travel time on the link a (h), $a \in G \cup R \cup W$
h_y	Handling time at the inland terminal s (h/TEU)
hh_y	Container storage time at the inland terminal s (h/TEU)
l_j	Customs clearance time at the seaport j (h/TEU)
m	Time value of container cargo ($/TEU)
u	Carbon tax ($/t$CO^2$)
Decision variables	
q_{ij}	Container volume from the city i to the seaport j
x_a	Number of containers on the link a
$t_a(x_a)$	Actual travel time on the link a, $a \in G \cup R \cup W$
f_k^{ij}	Number of containers on the path k connecting the city i and the seaport j
Z_{ijk}	Generalized transport costs when the city i and the seaport j are connected by the path k

Note: TEU = 20-ft equivalent unit; tCO^2 = ton carbon dioxide.

Model Specification

A port hinterland intermodal network has multiple features, such as mode change at rail or waterway terminals, capacity constraints, flow-dependent link performance functions, and the shipper's perceptual error of generalized transport costs. This section develops a model to capture these characteristics from the perspective of user equilibrium freight flows.

Define a transport network $G = (N, A)$, where N is the set of all nodes including inland cities, inland terminals, and seaports, and A is the set of all directed links including road links from cities to inland terminals, road links between cities, rail links, waterway links, processing links for inland terminal operation, and road links from inland terminals to seaports. A virtual node is added to each terminal node, and the link between the two nodes represents the process of container cargo operations at terminals. The notations used in this paper are displayed in Table 2.

Generalized Transport Costs

Shippers generally choose seaports according to the generalized transport costs between cities and seaports. Because of the influence of random factors, there is a difference between generalized transport costs perceived by the shipper, z_{ij}, and the actual transport costs, Z_{ij}. Thus, the perceived costs are the sum of the actual costs and the perceptual error term, ε_{ij}:

$$z_{ij} = Z_{ij} + \varepsilon_{ij}, \quad \forall i \in I; j \in J \tag{1}$$

The probability of the shipper i choosing the seaport j is as follows:

$$p_{ij} = p(z_{ij} < z_{ij'}), j, j' \in J, \text{且} j \neq j' \tag{2}$$

Random utility theory can describe the deviation between the shipper's perceived and actual transport costs, and the distribution of this deviation determines which model will be used. That is, the probability of a shipper choosing a seaport can be calculated from the actual costs and the error term. We assume that the error term follows the Gumbel distribution and obtain the following equation:

$$p_{ij} = \frac{\exp(-\theta Z_{ij})}{\sum_{j' \in J} \exp(-\theta Z_{ij'})} \tag{3}$$

Further, Z_{ij} and Z_{ijk} have the following relationship, and K_{ij} is different for each shipper because of the shipper's location and the set of modes:

$$Z_{ij} = \min(Z_{ijk \in K_{ij}}), \quad \forall i \in I; j \in J \tag{4}$$

When shippers choose seaports and paths, they always trade off the following components: inland direct transport costs, time costs, carbon emissions costs, and operational costs. In other words, the generalized transport costs for the path k, Z_{ijk} can be formulated as follows:

$$Z_{ijk} = Z^1_{ijk} + Z^2_{ijk} + Z^3_{ijk} + Z^4_{ijk}, \quad \forall i \in I; j \in J; k \in K_{ij} \tag{5}$$

The inland direct transport costs, Z^1_{ijk}, can be given by the sum of link transport costs along this path. We use b_a to denote transport costs for the link a, and α^{ij}_{ka} to represent the relationship between paths and links:

$$Z^1_{ijk} = \sum_{a \in G \cup R \cup W} b_a \alpha^{ij}_{ka}, \quad \forall i \in I; j \in J; k \in K_{ij} \tag{6}$$

The total transport time is composed of inland direct travel time, inland terminal handling time, storage time, and customs clearance time. The time costs, Z^2_{ijk}, are the product of the total time and the value of time (VOT) of containerized cargo:

$$Z^2_{ijk} = \left(\sum_{a \in (G \cup R \in W)} t_a(x_a) \alpha^{ij}_{ka} + \sum_{y \in Y} (h_y + hh_y) \alpha^{ij}_{ky} + l_j \right) \times m, \quad \forall i \in I; j \in J; k \in K_{ij} \tag{7}$$

where $t_a(x_a)$ is the actual travel time on the link a, h_y is the inland terminal handling time, hh_y is container storage time at the inland terminal, l_j is customs clearance time, and m is time value.

The inland direct travel time can be computed by the following functions, where β denotes penalty rate (*17*, *28*). When the road, rail, or waterway network is congested, the actual travel time depends on the joint effect of the number of containers transported by each shipper on this link and is the function of free-flow travel time (t^0_a) and link flow volume (x_a). Thus, shippers interact with each other. For example, if a shipper increases the number of containers transported on a link, the actual travel time and transport costs associated with that link will increase, leading to a change in the choice behavior of other shippers. The formulas imply that when the number of containers is close to transport capacity (C_a), the actual travel time for rail and waterway links is more sensitive to this factor than that for road links:

$$t_a(x_a) = t^0_a \left(1 + 0.15 \left(\frac{x_a}{C_a} \right)^{\beta} \right), \quad \forall a \in G \tag{8}$$

$$t_a(x_a) = t^0_a \left(1 + \left(\frac{x_a}{C_a} \right)^{\beta} \right), \quad \forall a \in R \cup W \tag{9}$$

The carbon emissions costs, Z^3_{ijk}, are mainly produced in the inland direct transport, terminal transshipment, and seaport operational processes:

$$Z^3_{ijk} = \left(\sum_{a \in G \cup R \cup W} e_a \alpha^{ij}_{ka} + \sum_{y \in Y} e_y \alpha^{ij}_{ky} + e_j \right) \times u / 1000, \quad \forall i \in I; j \in J; k \in K_{ij} \tag{10}$$

where e_y, e_a, and e_j represent the carbon emissions rate of the inland terminal, link, and seaport, respectively, and u is a carbon tax.

The operational costs, Z^4_{ijk}, which depend on handling costs at inland terminals (b_y) and operational costs at seaports (b_j), can be determined by the formula below:

$$Z^4_{ijk} = \sum_{y \in Y} b_y \alpha^{ij}_{ky} + b_j, \quad \forall i \in I; j \in J; k \in K_{ij} \tag{11}$$

Mathematical Model

If the port hinterland intermodal freight network achieves equilibrium status, the transport route, mode, and shippers' inland terminal choices fulfill the deterministic user equilibrium conditions, and seaport choice behavior is controlled by the logit formulation. At this point, shippers in each city allocate container freight volumes to each seaport based on a logit model, and the generalized transport costs of all selected paths connecting the same seaport and the same city are equal and not greater than that of the unused paths, which satisfies the Wardrop user equilibrium condition (*29*). The port

hinterland intermodal network equilibrium conditions can be expressed as the following model.

Minimize the objective function:

$$\begin{aligned} Z(f_k^{ij}, q_{ij}) = {} & m \sum_{a\in G\cup R\cup W} \int_0^{x_a} t_a(\omega)d\omega \\ & + \sum_{i\in I}\sum_{j\in J}\sum_{k\in K_{ij}}\sum_{y\in Y}(h_y + hh_y)\alpha_{ky}^{ij} \times m \times f_k^{ij} \\ & + \sum_{i\in I}\sum_{j\in J}\sum_{k\in K_{ij}} m \times l_j \times f_k^{ij} + \sum_{i\in I}\sum_{j\in J}\sum_{k\in K_{ij}} Z_{ijk}^1 f_k^{ij} \\ & + \sum_{i\in I}\sum_{j\in J}\sum_{k\in K_{ij}} Z_{ijk}^3 f_k^{ij} + \sum_{i\in I}\sum_{j\in J}\sum_{k\in K_{ij}} Z_{ijk}^4 f_k^{ij} \\ & + \frac{1}{\theta}\sum_{i\in I}\sum_{j\in J} q_{ij}(\ln q_{ij} - 1) \end{aligned} \tag{12}$$

The objective function contains seven parts: the first part is the sum of time costs during the gradual accumulation of containers on road links, rail links, and waterway links; the second part is the time costs of inland terminals; the third part is the time costs of seaports; the fourth, fifth, and sixth parts are inland direct transport costs, carbon emissions costs, and operational costs, respectively; the seventh part is the entropy-type function that represents the choice of seaport.

Subject to the following constraints:

$$x_a = \sum_{i\in I}\sum_{j\in J}\sum_{k\in K_{ij}} f_k^{ij}\alpha_{ka}^{ij}, \quad \forall a \in A \tag{13}$$

$$\sum_{j\in J} q_{ij} = P_i, \quad \forall i \in I \tag{14}$$

$$\sum_{k\in K_{ij}} f_k^{ij} = q_{ij}, \quad \forall i \in I; j \in J \tag{15}$$

$$x_a \leqslant C_a, \quad \forall a \in \overline{A} \tag{16}$$

$$f_k^{ij} \geqslant 0, \quad \forall i \in I; j \in J; k \in K_{ij} \tag{17}$$

$$q_{ij} \geqslant 0, \quad \forall i \in I; j \in J \tag{18}$$

Equation (13) means that the number of containers on the link is equal to the sum of the number of containers on all paths passing through this link; equation (14) represents the freight demand of each city that needs to be met; equation (15) ensures that the number of containers on all paths between the city and the seaport is equal to the container volume from the city to the seaport; equation (16) prevents flows on each link exceeding the capacity of that link; equations (17) and (18) are the value range of variables.

Proof of Equivalence

Next, we will prove the equivalence between the above model and the intermodal network equilibrium conditions. Setting λ_i, β_{ij}, η_a, ϖ_{ijk}, and υ_{ij} as the Lagrange multipliers of equations (14), (15), (16), (17), and (18), respectively, and constructing the generalized Lagrange function of the above model gives the following:

$$\begin{aligned} L = {} & Z\left(f_k^{ij}, q_{ij}\right) + \sum_{i\in I}\lambda_i\left(P_i - \sum_{j\in J} q_{ij}\right) \\ & + \sum_{i\in I}\sum_{j\in J}\beta_{ij}\left(q_{ij} - \sum_{k\in K_{ij}} f_k^{ij}\right) \\ & + \sum_{a\in\overline{A}}\eta_a\left(\sum_{i\in I}\sum_{j\in J}\sum_{k\in K_{ij}} f_k^{ij}\alpha_{ka}^{ij} - C_a\right) \\ & + \sum_{i\in I}\sum_{j\in J}\sum_{k\in K_{ij}}\varpi_{ijk}f_k^{ij} + \sum_{i\in I}\sum_{j\in J}\upsilon_{ij}q_{ij} \end{aligned} \tag{19}$$

By incorporating equations (12) and (13), the Karush–Kuhn–Tucker conditions of the above optimization model can be written as follows:

$$\begin{aligned} \frac{\partial L}{\partial f_k^{ij}} = {} & \frac{\partial Z\left(f_k^{ij}, q_{ij}\right)}{\partial f_k^{ij}} - \beta_{ij} + \sum_{a\in\overline{A}}\eta_a\alpha_{ka}^{ij} + \varpi_{ijk} \\ = {} & m \times \left(\sum_{a\in G\cup R\cup W} t_a(x_a)\alpha_{ka}^{ij} + \sum_{y\in Y}\left(h_y + hh_y\right)\alpha_{ky}^{ij} + l_j\right) \\ & + Z_{ijk}^1 + Z_{ijk}^3 + Z_{ijk}^4 - \beta_{ij} \\ & + \sum_{a\in\overline{A}}\eta_a\alpha_{ka}^{ij} + \varpi_{ijk} = 0, \quad \forall i \in I; j \in J; k \in K_{ij} \end{aligned} \tag{20}$$

$$\varpi_{ijk}f_k^{ij} = 0, \quad \forall i \in I; j \in J; k \in K_{ij} \tag{21}$$

$$\varpi_{ijk} \leqslant 0, \quad \forall i \in I; j \in J; k \in K_{ij} \tag{22}$$

$$\frac{\partial L}{\partial q_{ij}} = \frac{1}{\theta}\ln q_{ij} - \lambda_i + \beta_{ij} + \upsilon_{ij} = 0, \quad \forall i \in I; j \in J \tag{23}$$

$$\upsilon_{ij}q_{ij} = 0, \quad \forall i \in I; j \in J \tag{24}$$

$$\upsilon_{ij} \leqslant 0, \quad \forall i \in I; j \in J \tag{25}$$

$$\eta_a(x_a - C_a) = 0, \quad \forall a \in \overline{A} \tag{26}$$

$$\eta_a \geqslant 0, \quad \forall a \in \overline{A} \tag{27}$$

Note that the Lagrange multiplier, η_a, is associated with the link capacity, which means a penalty or impedance when the number of containers on the link exceeds the capacity. Alternatively, η_a can be considered as the delay costs caused by the link capacity constraint (*30*). Therefore, Z_{ijk} can be expressed according to the following formula:

$$Z_{ijk} = Z_{ijk}^1 + Z_{ijk}^2 + Z_{ijk}^3 + Z_{ijk}^4 + \sum_{a\in\overline{A}}\eta_a\alpha_{ka}^{ij} \tag{28}$$

The combination of equations (7), (20), and (28) yields the following:

$$Z_{ijk} - \beta_{ij} = -\varpi_{ijk}, \quad \forall i \in I; j \in J; k \in K_{ij} \tag{29}$$

From formulas (21) and (22) we can obtain the following:

$$Z_{ijk} - \beta_{ij} \geqslant 0, \ if\, f_k^{ij} = 0 \tag{30}$$

$$Z_{ijk} - \beta_{ij} = 0, \ if\, f_k^{ij} > 0 \tag{31}$$

Equations (30) and (31) term that β_{ij} is the minimum generalized transport costs between the city i and the seaport j. For each pair of cities and seaports, the generalized transport costs of all selected paths ($f_k^{ij} > 0$) are equal and not greater than that of the unused paths ($f_k^{ij} = 0$).

Similarly, according to equations (23) to (25), when $q_{ij} > 0$:

$$q_{ij} = e^{\theta(\lambda_i - \beta_{ij})}, \quad \forall i \in I; j \in J \tag{32}$$

Combining it with equations (4), (14), and (31):

$$e^{\theta\lambda_i} \sum_{j' \in J} e^{-\theta Z_{ij'}} = P_i, \quad \forall i \in I \tag{33}$$

If both sides of equation (33) are multiplied by $\frac{e^{-\theta Z_{ij}}}{\sum_{j' \in J} e^{-\theta Z_{ij'}}}$, equation (34) is consistent with the probability model of the shipper's choice of the seaport mentioned above:

$$q_{ij} = P_i \frac{e^{-\theta Z_{ij}}}{\sum_{j' \in J} e^{-\theta Z_{ij'}}}, \quad \forall i \in I; j \in J \tag{34}$$

Overall, in the proposed model, the shipper's behavior in choosing transport route, mode, and inland terminal conforms to the user equilibrium, and they choose seaports according to the logit model.

Solution Method and Algorithm

Note that the model takes into account capacity constraints of links and inland terminals; it is difficult to solve the problem directly using the conventional user equilibrium allocation method. This paper combines the augmented Lagrange multiplier method and the partial linearization method (*29*) to resolve the model. That is, capacity constraints are put into objective function by the augmented Lagrange multiplier method, thereby forming a sub-problem, which is resolved by the partial linearization method. The augmented Lagrange function can be expressed as follows:

$$L_\rho = Z(f_k^{ij}, q_{ij}) + \frac{1}{2\rho}\left\{\sum_{a \in \bar{A}} \max^2[0, \eta_a + \rho(x_a - C_a)] - \eta_a^2\right\} \tag{35}$$

where η_a is the Lagrange multiplier of equation (16), and ρ is the penalty constant. The sub-problem contains equations (13) to (15), (17), (18), and (35).

The algorithm for solving the problem is as follows:

Step 1: Setting initial values for η_a and ρ; number of iterations n is 1.

Step 2: Using the partial linearization method to settle the sub-problem; inland direct travel time function is $t_a(x_a) + \max\{0, {\eta_a}^n + \rho^n(x_a - C_a)\}$. Then, link flow ${x_a}^n$ and container volume ${q_{ij}}^n$ are generated.

Step 2.1: Initializing parameter; number of iterations m is equal to 1.

Step 2.2: Calculating the generalized transport costs ${Z_{ijk}}^m$ when the city i and the seaport j are connected by the path k.

Step 2.3: According to ${Z_{ijk}}^m$, attaining the minimum generalized transport costs ${Z_{ij}}^m$ between the city i and the seaport j.

Step 2.4: Adopting the logit model to identify container volume ${qq_{ij}}^m$ from the city i to the seaport j.

$$qq_{ij}{}^m = P_i \frac{e^{-\theta Z_{ij}{}^m}}{\sum_{j' \in J} e^{-\theta Z_{ij'}{}^m}}, \quad \forall i \in I; j \in J \tag{36}$$

Step 2.5: Assigning container flows ${qq_{ij}}^m$ to the path with the least generalized transport costs between the city i and the seaport j, and yielding the link flows ${xx_a}^m$.

Step 2.6: The number of containers on links and container flows from each city to seaports is updated by the successive weight average method (*31*).

$$x_a{}^{m+1} = x_a{}^m + \frac{n}{\sum_{i=1}^{n} i}(xx_a{}^m - x_a{}^m) \tag{37}$$

$$q_{ij}{}^{m+1} = q_{ij}{}^m + \frac{n}{\sum_{i=1}^{n} i}\left(qq_{ij}{}^m - q_{ij}{}^m\right) \tag{38}$$

Step 2.7: Judging algorithm termination condition. If $\tau \leqslant \zeta$ (ζ is the convergence criterion), go to Step 3; otherwise, $m = m + 1$, and go to Step 2.2.

$$\tau = \frac{\sqrt{\sum_{a \in A}\left(x_a{}^{m+1} - x_a{}^m\right)^2}}{\sum_{a \in A} x_a{}^m} + \frac{\sum_{i \in I}\sum_{j \in J} q_{ij}{}^m \left| q_{ij}^m - P_i \frac{e^{-\theta Z_{ij}{}^m}}{\sum_{j' \in J} e^{-\theta Z_{ij'}{}^m}} \right|}{\sum_{i \in I}\sum_{j \in J} q_{ij}{}^m} \tag{39}$$

Step 3: Updating the Lagrange multiplier and penalty constant.

$$\eta_a^{n+1} = \max\{0, \eta_a^n + \rho^n(x_a^n - C_a)\} \quad (40)$$

$$\rho^{n+1} = \begin{cases} \xi\rho^n, \sqrt{\sum_{a\in\bar{A}} \max^2(-\eta_a^n/\rho^n, x_a^n - C_a)} > \phi \\ \sqrt{\sum_{a\in\bar{A}} \max^2(-\eta_a^{n-1}/\rho^{n-1}, x_a^{n-1} - C_a)} \\ \rho^n, \sqrt{\sum_{a\in\bar{A}} \max^2(-\eta_a^n/\rho^n, x_a^n - C_a)} \leqslant \phi \\ \sqrt{\sum_{a\in\bar{A}} \max^2(-\eta_a^{n-1}/\rho^{n-1}, x_a^{n-1} - C_a)} \end{cases} \quad (41)$$

Step 4: If $\max_{a\in\bar{A},\, \eta_a^n>0} \left\{\frac{|x_a^n - C_a|}{C_a}\right\} \leqslant \zeta\zeta$ ($\zeta\zeta$ is the convergence criterion), stop the iteration; otherwise, $n = n + 1$, and go to Step 2.

An Application of the Model

Case Study Description

The Yangtze River Economic Belt (YREB) in China involves 11 provinces and municipalities: Shanghai, Jiangsu, Zhejiang, Anhui, Jiangxi, Hubei, Hunan, Chongqing, Sichuan, Yunnan, and Guizhou. It covers an area of approximately 2.05 million square kilometers, contains more than 40% of the country's population, and contributes nearly 45% of the GDP. Container cargoes in the YREB region are mainly imported and exported through the Port of Shanghai and the Port of Ningbo-Zhoushan, which are two of the top container ports in China. The overview of the port hinterland intermodal freight network for YREB is presented in Figure 1, and it comprises 37 inland cities, two seaports, nine inland waterway terminals, and 10 inland rail terminals. Inland terminals charge shippers according to guidelines established by the government, and fees do not vary with flow volume. Import and export containers at inland terminals have different storage areas and operational processes, and the mutual influence is small. This paper assumes that imports and exports are two processes that do not interfere with one another. The export value of each province and municipality is twice the import value, so this paper only pays attention to export freight. Import and export flows will be considered simultaneously in future research work. The choice of inland cities is based on export values. The top five cities in each province are selected with the exception of Yunnan and Guizhou, where two municipalities are directly selected as shown in Table 3. For road transport, inland cities, terminals, and seaports are connected by the Shanghai–Chengdu expressway, Nanjing–Hangzhou expressway, Hangzhou–Changsha expressway, Shanghai–Hangzhou–Ningbo expressway, and the Huangshan–Hangzhou expressway. The container inland waterway terminal system has gradually developed, and it mainly includes the ports of Taicang

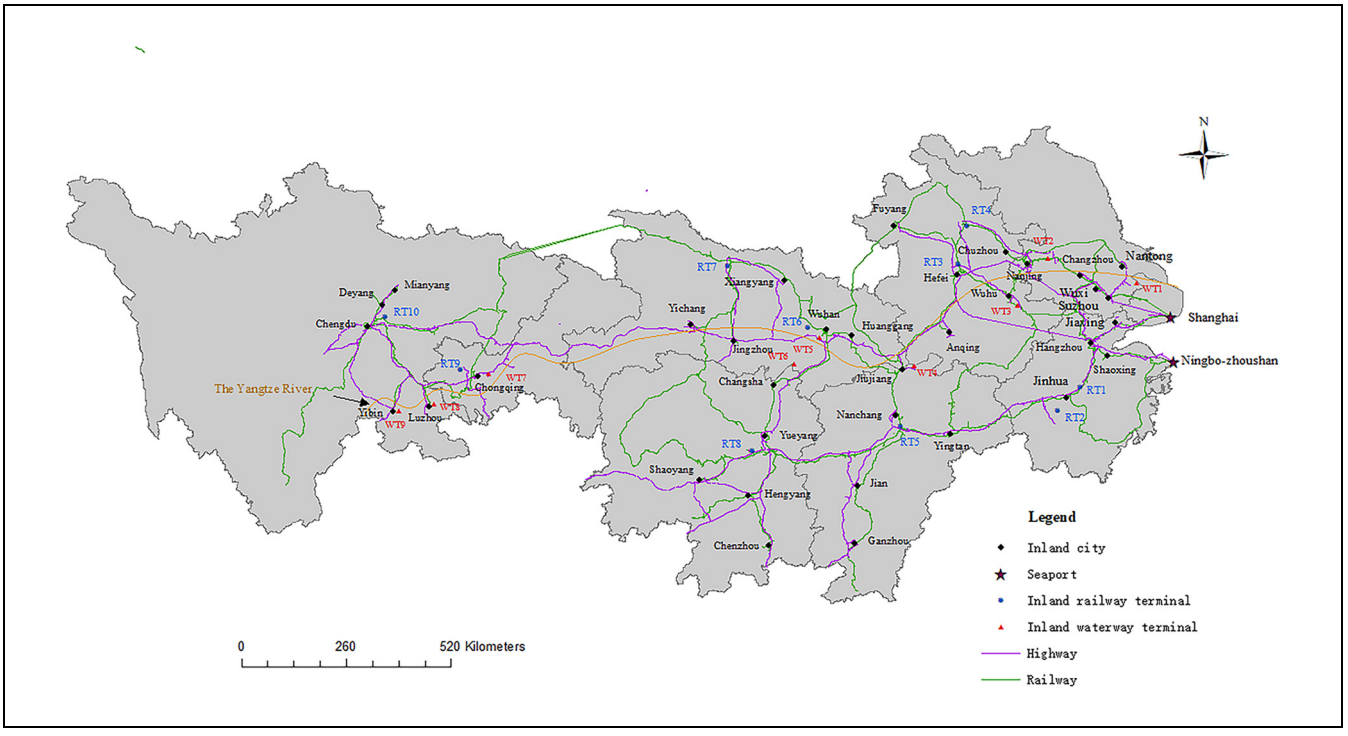


Figure 1. Intermodal freight network for the Yangtze River Economic Belt.

Table 3. Selected Inland Cities and Export Values

City name	Province or municipality	Export value (billion dollars)	City name	Province or municipality	Export value (billion dollars)
Chengdu	Sichuan	59.23	Hangzhou	Zhejiang	53.39
Mianyang	Sichuan	1.13	Jiaxing	Zhejiang	32.86
Yibin	Sichuan	1.79	Shaoxing	Zhejiang	34.49
Deyang	Sichuan	1.19	Suzhou	Jiangsu	186.87
Luzhou	Sichuan	0.65	Nanjing	Jiangsu	49.11
Chongqing	Chongqing	60.45	Wuxi	Jiangsu	51.24
Wuhan	Hubei	20.52	Nantong	Jiangsu	25.88
Xiangyang	Hubei	2.85	Changzhou	Jiangsu	25.94
Yichang	Hubei	2.66	Changsha	Hunan	22.36
Jingzhou	Hubei	1.49	Yueyang	Hunan	2.67
Huanggang	Hubei	0.86	Chenzhou	Hunan	3.34
Hefei	Anhui	22.79	Hengyang	Hunan	2.59
Wuhu	Anhui	4.87	Shaoyang	Hunan	3.72
Chuzhou	Anhui	2.92	Nanchang	Jiangxi	10.29
Fuyang	Anhui	1.21	Jian	Jiangxi	5.65
Anqing	Anhui	1.49	Ganzhou	Jiangxi	6.05
Shanghai	Shanghai	198	Jiujiang	Jiangxi	5.37
Ningbo	Zhejiang	92.63	Yingtan	Jiangxi	1.56
Jinhua	Zhejiang	66.69			

(WT1), Nanjing (WT2), Wuhu (WT3), Jiujiang (WT4), Wuhan (WT5), Yueyang (WT6), Chongqing (WT7), Luzhou (WT8), and Yibin (WT9) (see Figure 2), and ships from these terminals all sail through the Yangtze River to seaports. With national investment in infrastructure, inland rail terminals with higher operational capability include Yiwu (RT1), Jinhua (RT2), Hefei (RT3), Bengbu (RT4), Nanchang (RT5), Wuhan (RT6), Xiangyang (RT7), Changsha (RT8), Chongqing (RT9), and Chengdu (RT10) railway stations (see Figure 3).

The model is exploited to assess the impact of various transport policies on the port hinterland intermodal freight network, from the perspective of carbon emissions, mode shift, and network costs.

Input Data

Container freight transport demand is obtained through multiplying the export value by the seaborne export ratio and dividing it by the cargo value. On the basis of the preliminary survey, the cargo value is 38880 $/TEU, the seaborne export ratio of Shanghai, Jiangsu, Zhejiang, and Anhui is 0.7, and others are 0.3. The model proposed in this paper applies to data for any period, for example, daily, weekly, monthly, and yearly. Because of the lack of daily, weekly, and monthly data on freight demand and capacity, annual data are used in the case study. When both flow and capacity in equations (8) and (9) are divided by the same value, the calculation result remains unchanged. The carbon emissions rate, direct transport costs, operational costs, and other time variables for a single container have no relationship with freight demand and capacity. Therefore, the period of data used does not affect the conclusion of the transport policy analysis. The road distance was obtained from the Baidu Map, rail distance and cost data were obtained from China Railway's website, and waterway distance was acquired from the Yangtze River Waterway Bureau's website. Ships from the ports of Chongqing, Luzhou, and Yibin sailing to seaports need to pass the Three Gorges Dam, which takes approximately 72 h, including ship safety inspections and ship waiting time. Furthermore, the penalty rate β is 4, and the deviation parameter θ is 2. Container storage time at inland waterway and rail terminals is 72 h and 48 h, respectively, and the VOT is 6.49 $/TEU/h (*32*). VOT is related to the type of cargo, and the value for VOT in this paper is average. Likewise, costs and carbon emissions data are estimated in respect of previous research (*22*, *33*), and seaport information has been obtained from ports' official websites and statistical yearbooks, as shown in Table 4. Finally, shippers in each province and municipality can select the same set of modes, which is determined by road, inland rail terminal and inland waterway terminal.

Another important parameter is the capacity to transport containers within the planning period. Because the planning cycle of this case study is one year, the inland terminal processing capacity is considered to be the maximum design capacity from the associated official website. The highway is supposed to be eight lanes in both directions, and with respect to other cars driving on the road, its capacity for trucks is 9,125,000 per year. In China, waterway vessels may call at multiple terminals, whereas the railway transports containers to seaports

Table 4. The Main Input Parameters of Cost and Time

Parameters	Transport mode		
	Road	Waterway	Rail
Transport costs			
Base price ($/TEU)	0	0	68
Average cost ($/TEU/km)	1.42	0.17	0.49
Speed (km/h)	120	18	80
Carbon emissions rate (kg/TEU/km)	2.19	0.11	0.42

Parameters	Inland terminal	
	Waterway	Rail
Handling costs ($/TEU)	38.64	38.64
Handling time (h/TEU)	1.50	1.50
Carbon emissions rate (kg/TEU)	5.80	5.80
Drayage fee		
Base price ($/TEU)	46.37	46.37
Average cost ($/TEU/km)	3.09	3.09

Parameters	Seaport	
	Shanghai	Ningbo-Zhoushan
Customs clearance time (h/TEU)	1.86	2.68
Port operational costs ($/TEU)	100.89	108.78
Carbon emissions rate (kg/TEU)	3.20	5.80
Drayage fee ($/TEU)	37.09	10.82

Note: TEU = 20-ft equivalent unit.

directly in most cases. Indirect transport data is difficult to investigate. Therefore, the paper only considers direct transport capacity for rail and waterway links. Because some rail terminals now open routes to one seaport only, and capacity and routes are gradually increasing, we assume that each inland rail terminal has four trains per day to each seaport, and one train can transport 100 TEUs. inland Waterway transport capacity varies among inland waterway terminals; the route density is two to three times per day, and each ship may load around 300 TEUs.

Model Validation

A validation test is required to check the quality of the model. Because some rail terminals temporarily only open railway lines to one seaport and transport capacity is gradually increasing, the verification model is carried out by simulating the status quo of the YREB, ignoring carbon emissions costs, and adjusting the rail transport capacity according to the actual situation. Based on the available observations, the results of the verification model are evaluated at the inland rail and inland waterway terminal level, by comparing the modeled traffic volume with the actual traffic volume, as shown in Figures 2 and 3. The coefficient of determination R^2 between the calculated value and the measured value for the rail terminal level and waterway terminal level is 0.87 and 0.81, respectively. From this point of view, the model fit is satisfactory.

Transport Policy Analysis

Five scenarios are implemented to assess transport policies. The carbon tax ranges from 0 to 1,000 in increments of 10 $/tCO^2, and the subsidy is 0 to 700 in increments of 10 $/TEU. Because the inland terminal processing capacity is far greater than the mode transport capacity, expansion of the latter is another policy measure, which is expressed as a percentage increase of existing capacity, with the value ranging from 0% to 200% in increments of 20%.

Scenario 1: Carbon Tax Policy. The carbon tax internalizes the external environmental costs, and Figure 4 shows the relationship between total network costs and the carbon tax. Internal costs gradually increase and a jump occurs when the carbon tax is about 850 $/tCO^2. It is obvious that costs saved by replacing road transport with rail or inland waterway transport cannot make up for the additional transport costs. The reason for this is that the time required for road transport is shorter and the drayage

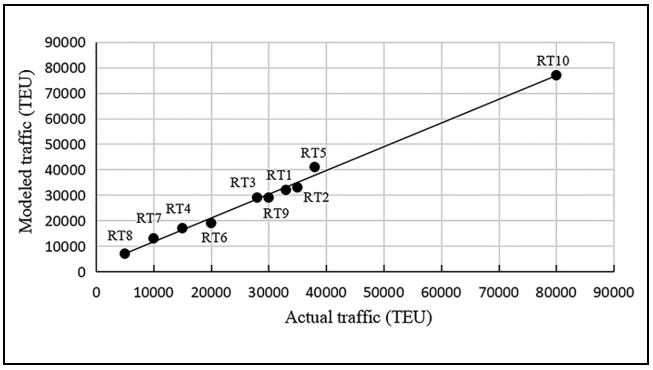


Figure 2. Calculation results of rail terminal.
Note: TEU = 20-ft equivalent unit; RT1 = Yiwu; RT2 = Jinhua; RT3 = Hefei; RT4 = Bengbu; RT5 = Nanchang; RT6 = Wuhan; RT7 = Xiangyang; RT8 = Changsha; RT9 = Chongqing; RT10 = Chengdu.

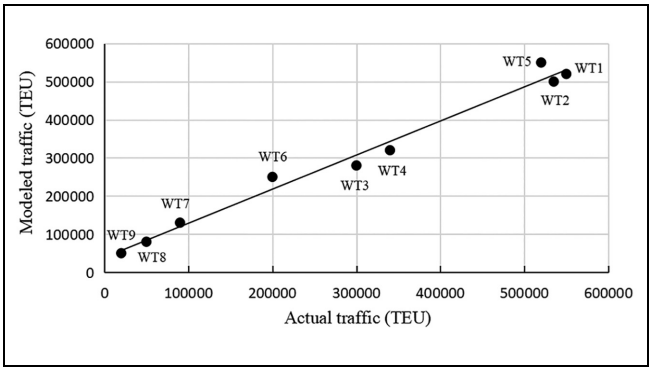


Figure 3. Calculation results of waterway terminal.
Note: TEU = 20-ft equivalent unit; WT1 = Taicang; WT2 = Nanjing; WT3 = Wuhu; WT4 = Jiujiang; WT5 = Wuhan; WT6 = Yueyang; WT7 = Chongqing; WT8 = Luzhou; WT9 = Yibin.

fee required is zero compared with rail or inland waterway transport. The changing trend of total network costs is consistent with internal costs, and external costs have less impact as a result of the small proportion of external costs in total network costs.

It can be seen from Figure 5 that the total market share of rail and inland waterway transport improves, with the carbon tax increasing. This implies the carbon tax can promote mode shift from road transport to rail and inland waterway transport. However, with a fairly high carbon tax, the road transport share does not decline at a significant rate because the capacity to transport containers by rail or inland waterway has an upper limit, and carbon emissions costs are just a small part of generalized transport costs. Interestingly, the rail transport share reaches its maximum at a carbon tax of 290 $/tCO^2; it then decreases and the inland waterway transport share increases. This proves that more container flows are shifted to inland waterway routes with lower emissions when the carbon tax increases.

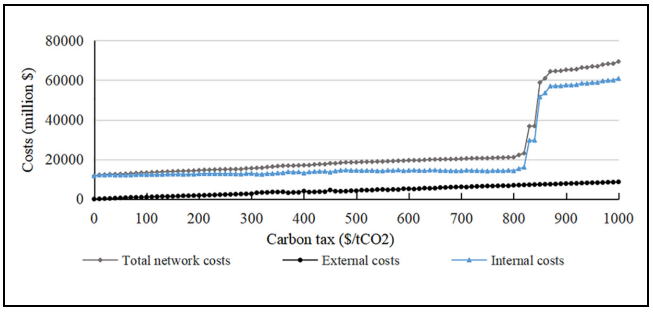


Figure 4. The relationship between total network costs and the carbon tax.
Note: tCO^2 = ton carbon dioxide.

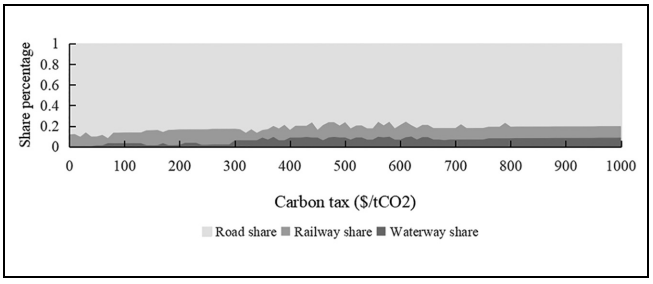


Figure 5. The relationship between mode market share and the carbon tax.
Note: tCO^2 = ton carbon dioxide.

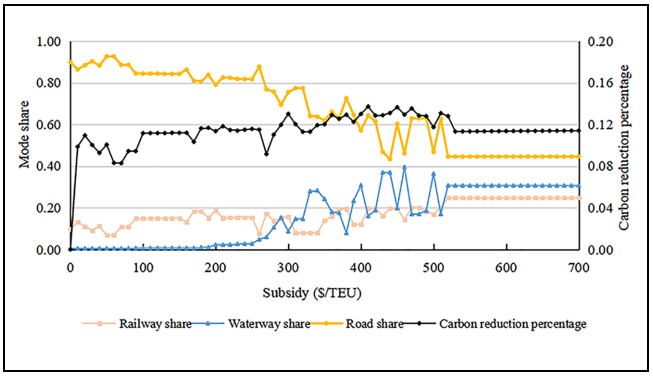


Figure 6. Mode market share and carbon emissions with changes in subsidy.
Note: TEU = 20-ft equivalent unit.

Scenario 2: Intermodal Transport Subsidy. Intermodal subsidies are monetary incentives provided by the government to the corresponding shipper if a container is transported to a seaport by rail or inland waterway. Figure 6 illustrates the relationship between the subsidy and the mode market share/carbon emissions. The road transport share varies mainly between 43% and 92%, which indicates the subsidy is important for reducing road transport volume.

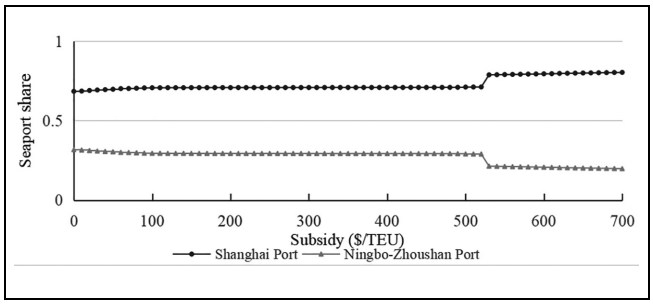


Figure 7. Share of different seaports.
Note: TEU = 20-ft equivalent unit.

However, in most cases, when subsidies exceed 260 $/TEU, they have a greater impact on inland waterway transport than on rail transport. Clearly, the percentage of emissions reduction fluctuates between 8% and 14% and there is no observed linear trend of increase or decrease. Two primary factors may bring about such a pattern of emissions reduction. First, the subsidy drives the mode shift from road transport to rail or inland waterway transport for cities with only two modes, but may promote the mode shift from inland waterway transport to rail transport for cities with three modes. Second, under the limitation of transport capacity, in that one city receives a subsidy, this may induce shippers in other cities to choose road transport because of changes in generalized transport costs.

With regard to container flows to seaports, Figure 7 shows the relationship between seaport volume and subsidy. The Port of Shanghai is more competitive than the Port of Ningbo-Zhoushan; therefore, the number of containers transported to the Port of Shanghai is more than the Port of Ningbo-Zhoushan receives. If the goal is to enhance the Port of Ningbo-Zhoushan's share effectively, a higher subsidy than that given to the Port of Shanghai should be provided or a subsidy scheme developed for each city according to the actual situation.

Scenario 3: Carbon Tax and Capacity Expansion. The network performances for 1,111 situations are shown in Figure 8. Although there are some ups and downs, total carbon emissions generally show a downward trend with an increase in carbon tax when transport capacity remains stable. In most situations, total emissions decrease as transport capacity increases, because more containers can be transported by rail or inland waterway. However, in some situations, higher capacity does not lower total carbon emissions. The reason can be clarified from flow results, that is, when rail capacity improves, more containers are shifted from inland waterway transport to rail transport because of smaller generalized costs. More importantly, combining the two policies can be more effective in reducing carbon emissions than a single carbon tax policy, especially when the value of carbon tax is low.

Scenario 4: Intermodal Transport Subsidy and Capacity Expansion. Figure 9 shows how the total market share of rail and inland waterway transport varies with subsidy and transport capacity, with some dots almost overlapping, for example, the base situation and a 20% increase

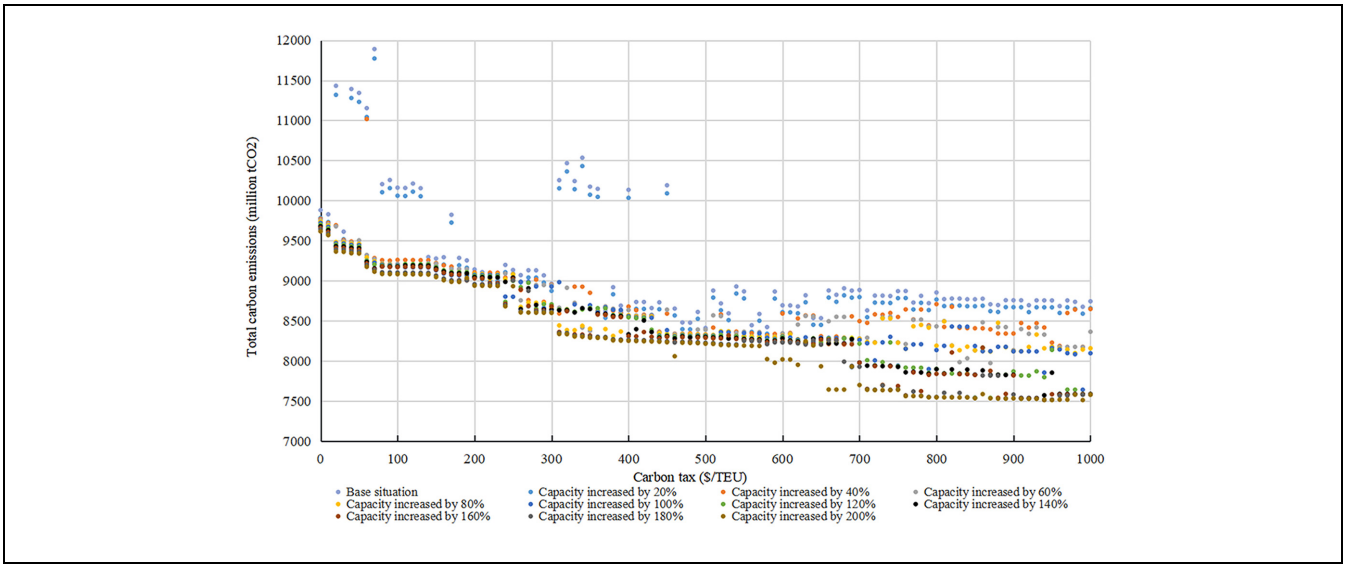


Figure 8. Carbon emissions for the carbon tax and capacity expansion scenarios.
Note: tCO^2 = ton carbon dioxide; TEU = 20-ft equivalent unit.

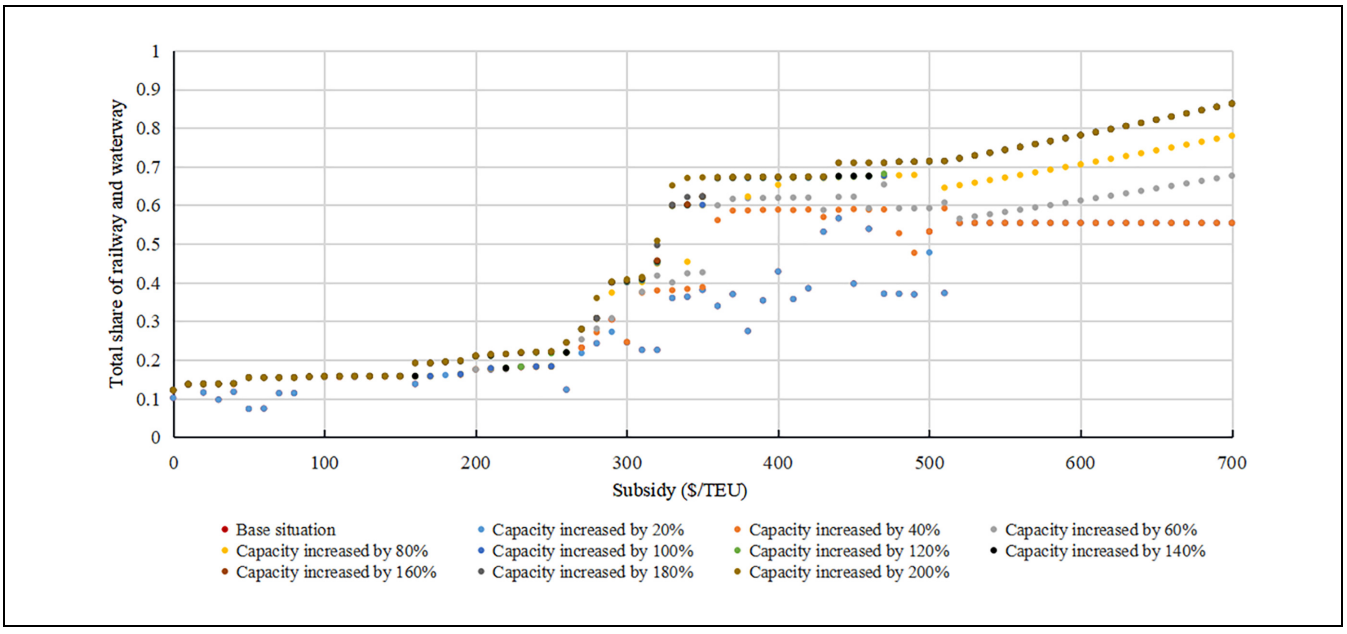


Figure 9. Total market share of rail and inland waterway transport for the subsidy and capacity expansion scenarios.
Note: TEU = 20-ft equivalent unit.

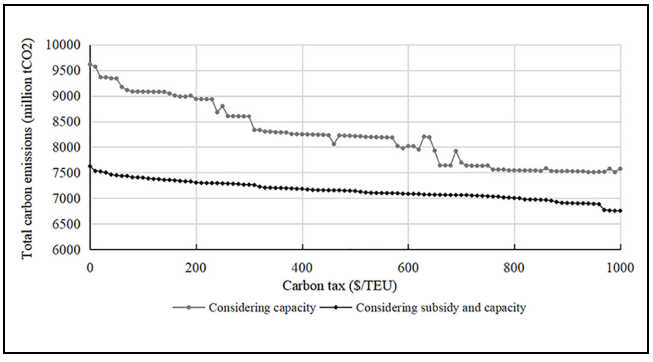


Figure 10. Optimal carbon emissions under different scenarios.
Note: tCO^2 = ton carbon dioxide; TEU = 20-ft equivalent unit.

in capacity. As expected, the total share is at its largest only when both the subsidy and capacity reach the maximum. This is reasonable, because the subsidy can reduce rail and inland waterway transport costs to attract shippers to choose these modes, and increasing capacity allows these modes to transport more containers during the planning period. The changes in total share contrasted with the base situation also confirm that considering subsidy and capacity expansion policies simultaneously plays a more important role in raising the share of eco-friendly modes than looking at them separately.

Scenario 5: Carbon Tax, Intermodal Transport Subsidy, and Capacity Expansion. Figure 10 shows the total carbon emissions for the best situation when varying the carbon tax. One curve illustrates the best situation, which is the optimal transport capacity for a carbon tax ranging from 0 to 1,000 $/$tCO^2$. Another curve shows the best situation in which carbon tax, intermodal subsidy, and capacity expansion policies are all available. The comparison of the two curves reveals that combining the three policies will reduce carbon emissions more effectively than other scenarios.

Moreover, the capacity curve shows total carbon emissions sometimes increase when the carbon tax gets bigger, for example, when the carbon tax is 680 and 690 $/$tCO^2$. The reason is that the carbon tax influences container volume to seaports because of different carbon emissions rates here, and there are differences in carbon emissions from each city to various seaports. Moreover, the gap between the two curves tends to decrease, suggesting that the effect of subsidies on reducing carbon emissions has depreciated significantly with a larger carbon tax.

Sensitivity Analysis

Sensitivity analyses are performed on cost, time, carbon emissions, and θ to examine the robustness of the proposed model, and they can also explore the impact of these changes on port hinterland freight planning.

Cost Parameters. Cost parameters include VOT, and the average cost of road, rail, and inland waterway transport. VOT changes with the type of cargo; the average cost of the above modes depends on the carrier, and the

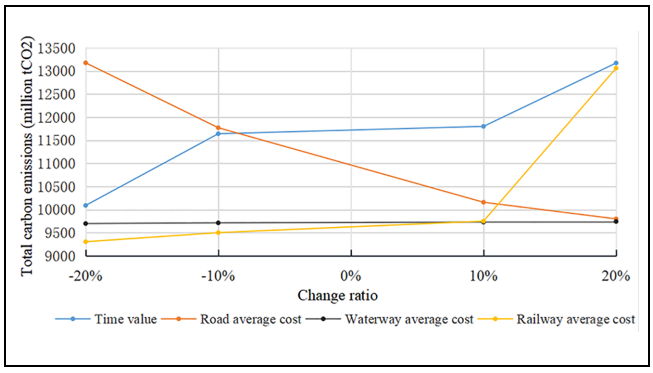


Figure 11. The relationship between total carbon emissions and cost parameters.
Note: tCO^2 = ton carbon dioxide.

factors leading to variation in the average cost may include operational costs, loading and unloading, supply and demand, business competition, and so forth. This paper only analyzes the impact of cost parameters on port hinterland freight, and does not consider the reasons for changes in the various parameters. Figure 11 gives the results of total carbon emissions for various situations. The average cost of road transport is the parameter with the largest impact on carbon emissions, with a difference of 34% when the cost increases and decreases by 10%. In addition, total emissions increase as VOT becomes bigger, because more time is required to complete journeys by rail and waterway than by road, and higher VOT will cut down the competitiveness of these two modes. The comparison of the rail and inland waterway transport curves might lead to the conclusion that if average rail transport costs increase by over 10%, flows on rail links will be shifted to road links, and average inland waterway transport costs have little impact on emissions.

Time Parameters. Time parameters refer indirectly to free-flow speeds of all modes; the speeds affect the actual travel time from inland cities to seaports. Figure 12 indicates the relationship between the market share of the mode and the free-flow speed. Obviously, the rail transport market share is most sensitive to rail speed. On the other hand, the network behavior has only slight sensitivity to waterway speed and road speed.

Carbon Emissions Parameters. This section analyzes the effects of carbon emissions rate on total emissions, as shown in Figure 13, in which each bar represents the total carbon emissions when changing the emissions rate of one mode. It testifies that total carbon emissions are less sensitive to rail and inland waterway transport because the carbon emissions rate of the two modes is relatively small. In contrast, total carbon emissions are closely related to road transport carbon emissions. Figure 13 indicates that for the YREB freight network, a reduction in carbon emissions can be obtained from measures that assist in bringing down road transport emissions, such as popularizing clean energy freight vehicles and using information technology to avoid trucks waiting in logistics parks.

Deviation Parameter θ. According to the previous analysis, the larger the value pf θ, the more the shipper's perception of the generalized transport costs aligns with the actual situation and the less the uncertainty. Figure 14 shows the container volume to each seaport for different parameter value levels and compares them with the base situation in which there is no generalized costs perceptual error. It can be seen that when θ is 2 and 10, container volume results are different from the base situation, whereas, if the parameter grows to 1,000, the flow allocation result is almost equal to the base situation. Taking into account the implications of the deviation parameter mentioned earlier, the results obtained as the variable changes are reasonable. This clearly indicates that the random factors in the network will continue to decrease as the accuracy of the shipper's perception of generalized transport costs increases. Indeed, this also verifies the correctness of the model.

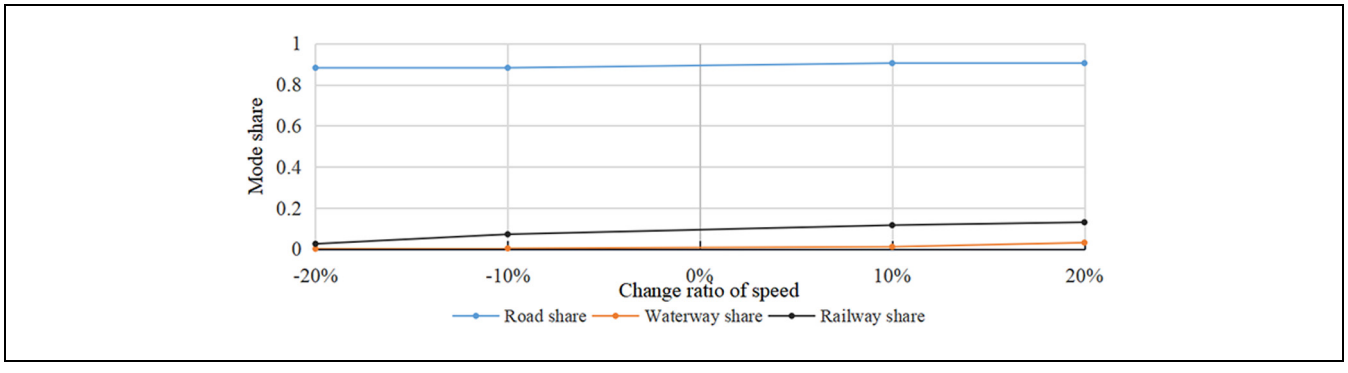


Figure 12. The relationship between mode market share and mode speed.

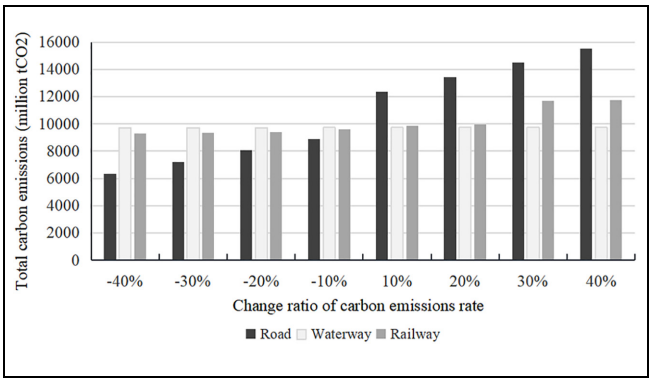


Figure 13. The relationship between total carbon emissions and emissions rate.
Note: tCO^2 = ton carbon dioxide.

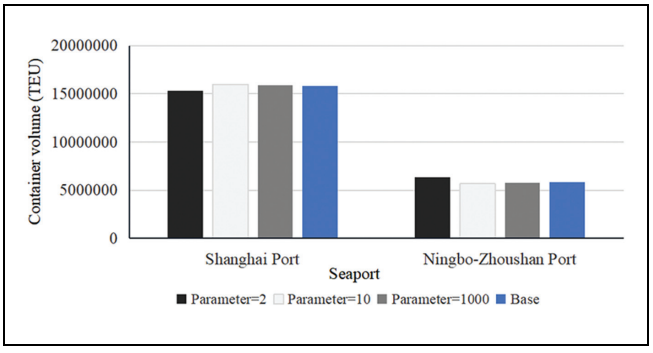


Figure 14. Container volume to seaports when changing deviation parameter.
Note: TEU = 20-ft equivalent unit.

Conclusion

This paper presents a generic model of the port hinterland intermodal freight network, which simultaneously considers shippers' choices of transport route, mode, inland terminal, and port, as well as constraints on transport and inland terminal processing capacity. To capture network characteristics, such as flow-dependent link performance functions and the shipper's perceptual error of generalized transport costs, the proposed model presumes that shippers choose route, mode, and inland terminals in a user equilibrium manner, and that container flows split between seaports follow the logit formulation. In addition, an augmented Lagrange multiplier method combined with the partial linearization method is designed to resolve the problem. Furthermore, the model and algorithm are used for a case study of hinterland transport in China to gain insight into the effects of transport policies on carbon emissions, mode shift, and network costs. Finally, sensitivity analyses on cost, time, carbon emissions, and the deviation parameter are conducted via simulation experiments. The main conclusions are summarized as follows:

(1) Charging the rate of carbon tax not only promotes mode shift from road to rail and inland waterway transport, but also increases total network costs. The costs saved by replacing road transport with rail or inland waterway transport cannot cover the additional time costs and drayage fees. Thus, the additional costs can significantly increase generalized transport costs for shippers, especially when the carbon tax is at a high rate. Furthermore, the competitiveness of inland waterway transport continuously improves with an increasing carbon tax.
(2) An intermodal transport subsidy can reduce road transport market share, and inland waterway transport market share is more sensitive to being offered a subsidy than rail transport. If the goal is to reduce carbon emissions, a subsidy of 290 $/TEU is recommended, because the percentage of emissions reduction at this time is close to the maximum. In addition, expanding mode transport capacity and providing subsidies may lead to an increase in total carbon emissions, which is contrary to what is expected. Consequently, the government should act cautiously to prevent the paradoxical phenomenon when designing transport policies.
(3) Limited mode transport capacity reduces the effects of introducing a carbon tax and offering intermodal transport subsidies, but a combination of all three policies will be more effective than separate policies in reducing carbon emissions and increasing the market share of rail and inland waterway transport. In addition, with regard to the target network performances, the combined policy provides more options for optimizing the port hinterland intermodal freight network.
(4) The results of the sensitivity analysis are revealing. The carbon emissions rate of road transport and the time taken for goods to be transported via rail and inland waterway have an important influence on total carbon emissions. Therefore, two measures offer the possibility of reducing carbon emissions: one is to use new technologies to improve truck energy efficiency and develop clean energy; the other is to optimize the intermodal transport operational process to shorten transport time and increase efficiency.

There are some study limitations in this paper. The shippers are homogeneous, and only export containers are considered; import containers and empty containers

are excluded. There are many variables used in the model, and the values of some of these deviate somewhat from the actual values because of the availability of data. Thus, three directions for future research are suggested. First, it is important to integrate shipper heterogeneity into the model, because shippers dealing with high VOT are willing to choose a faster mode of transport. Because the total investment amount for the port hinterland freight system in China is limited, the second direction worth investigating is extending the model. The intention here would be to recommend how to optimize the allocation of a given budget to develop construction plans or subsidy schemes that minimize the environmental impact and/or maximize mode shift from road transport to eco-friendly modes, considering transport in relation to both imports and exports, and empty containers too. Third, the research on port hinterland intermodal freight networks requires a lot of data, and exploring how to obtain more accurate data through conducting surveys could be a further direction for future research.

Author Contributions

The authors confirm their contribution to the paper as follows: study conception and design: Yunqiang Wu, Rong Zhang; data collection: Yunqiang Wu; analysis and interpretation of results: Yunqiang Wu, Rong Zhang; draft manuscript preparation: Yunqiang Wu, Rong Zhang. All authors reviewed the results and approved the final version of the manuscript.

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